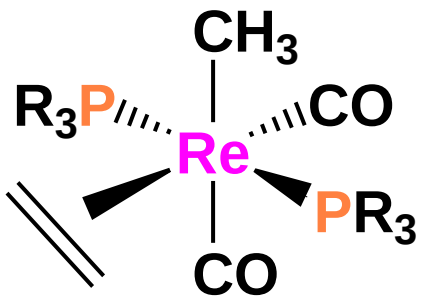
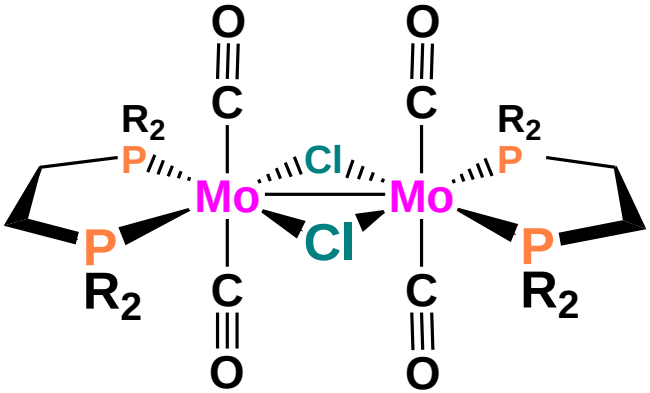
Transition metal organometallic compounds & Catalysis

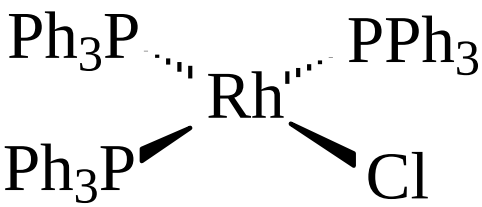
French Chemist L. C. Cadet 1760 As_2Me_4 dicacodyl

Which one is organometallic? $Ni(CO)_4$ or NaCN ?

Metal-carbon bond: a few from many?







What is catalyst:
A substance that increases the rate of a chemical reaction without itself undergoing any permanent chemical change.

What is catalysis: Catalysis is the process of modifying a chemical reaction with the use of a catalyst.

Turnover Number:
The **number of moles** of substrate that **a mole** of catalyst can convert before becoming inactivated.
TON = moles of product/moles of catalyst

Turnover Frequency:
The **number of moles** of substrate that **a mole** of catalyst can convert per unit time.
TOF = TON/time

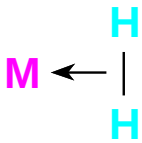
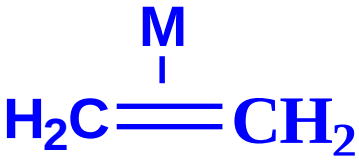
Selectivity : Ratio of substrate converted to desired product to total substrate converted

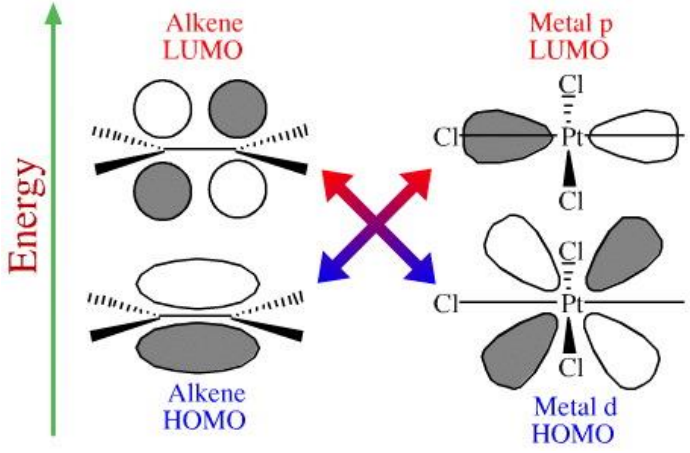
Catalytic cycle : cycle of reactions that comprise the mechanism, always returning to its original form

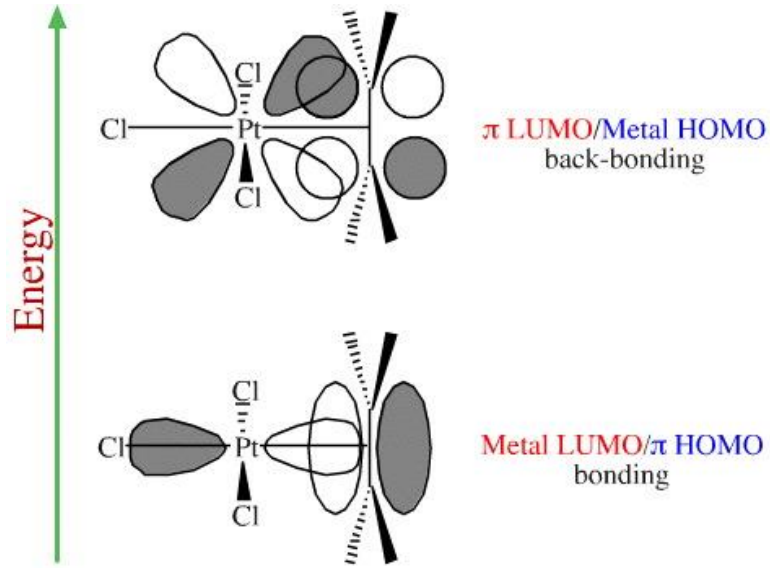
Yield: Mass of product actually formed/ mass of product that could be made

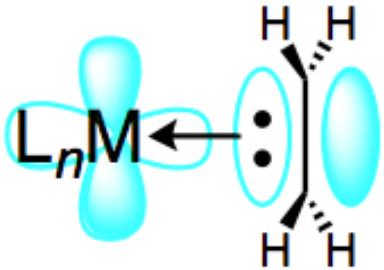
Atom economy: mass of atoms/total mass of atoms in reactants
Molecular wt. of desired product/molecular wt of all reactants

Organometallic Compound: Looking closer

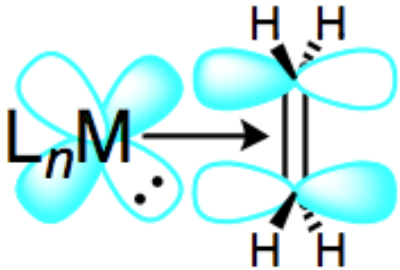
Ligand Name	Bonding Type
Molecular Hydrogen: H_2	
Hydride H^-	M-H
Phosphine: PR_3	M-PR_3
Carbonyl: C≡O	M−C≡O
Alkyl , Aryl	M-CR M-Ph
Alkene	



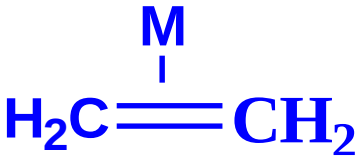




Filled π orbital
Empty *d* orbital



Filled metallic HOMO
Empty π* orbital



How can you change the rate of reaction???

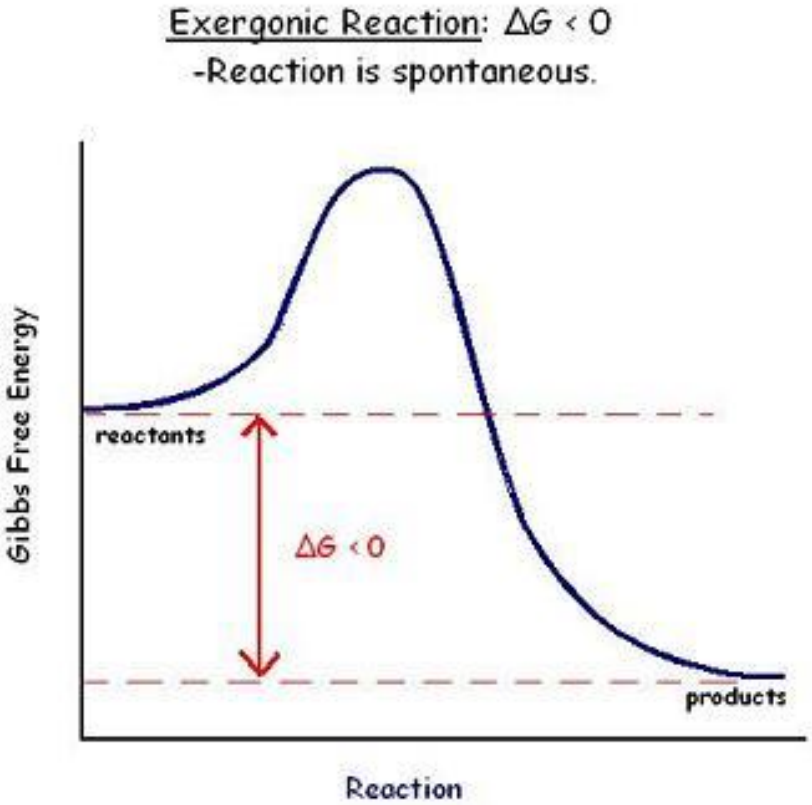
- Increasing the temperature of the reaction

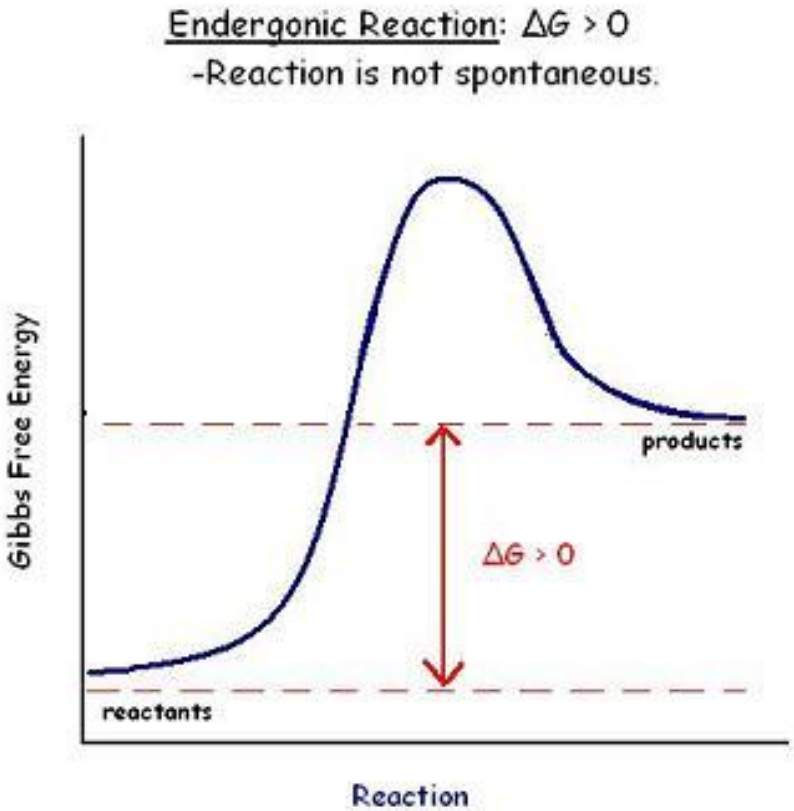
Problems

Expensive

Induce side-reaction

- By using some catalyst


Exergonic Reaction: ΔG < 0
-Reaction is spontaneous.
Gibbs Free Energy
reactants
ΔG < 0
products
Reaction


Endergonic Reaction: ΔG > 0
-Reaction is not spontaneous.
Gibbs Free Energy
products
ΔG > 0
reactants
Reaction

google

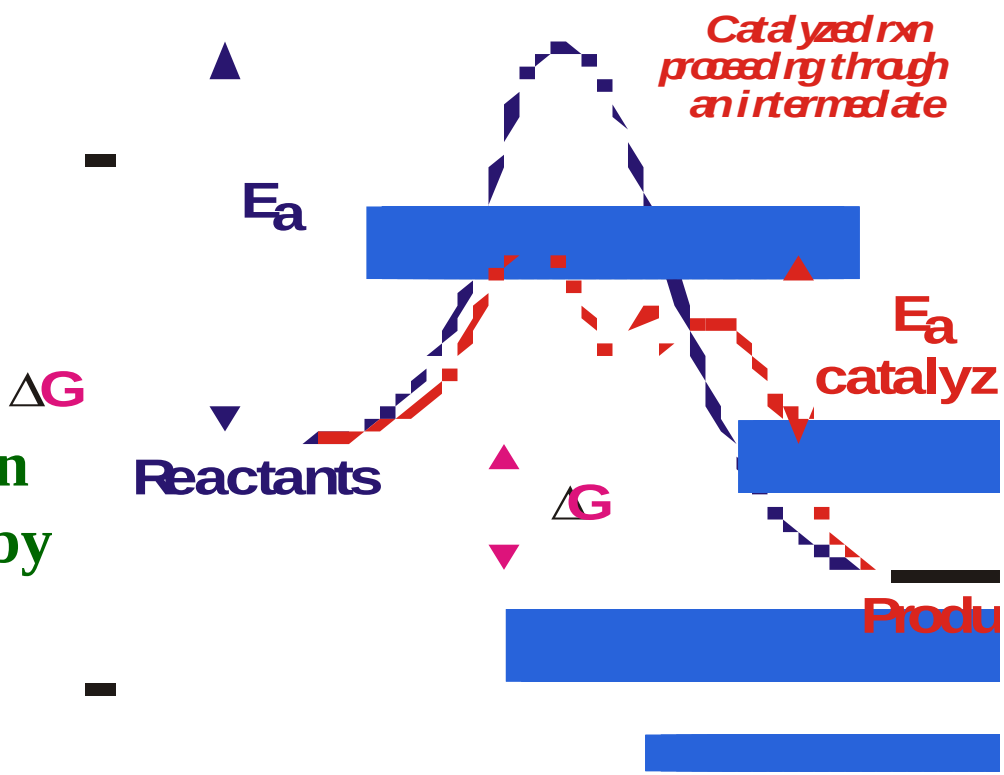
Catalysis : Basics

$$A + B \xrightarrow{\text{Catalyst}} C$$

Heterogeneous

Homogeneous

A catalyst lowers the activation barrier for a transformation, by introducing a new reaction pathway.



It does not change the thermodynamics!!

Catalysis : Why

Synthesis of chemicals… pharmaceutical, agricultural

Catalytic converter … environmental

Biological system – efficient catalyst

Homogeneous and Heterogeneous

How to select a good catalyst?

Activity: Related to rate of reaction (also called turnover)

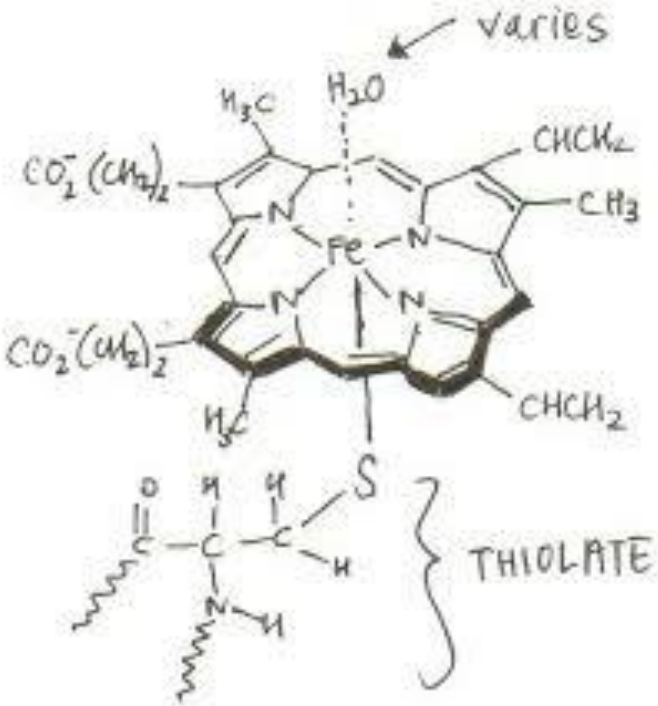
Efficiency: Good activity

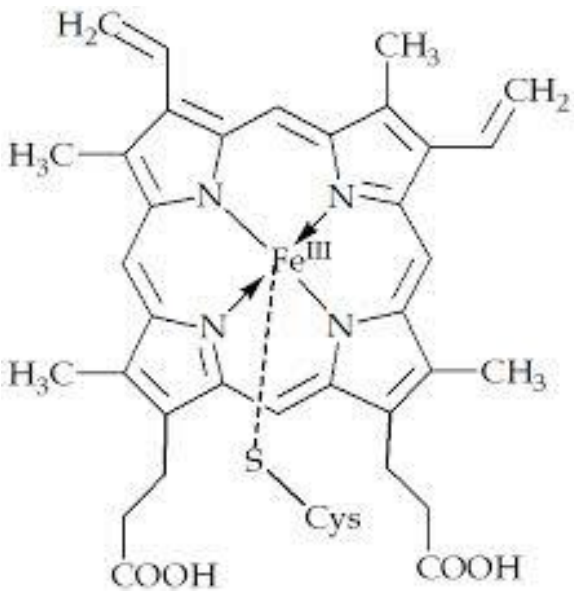
Selectivity: Byproducts should be minimized

Lifetime: It is costly to replace the catalyst frequently

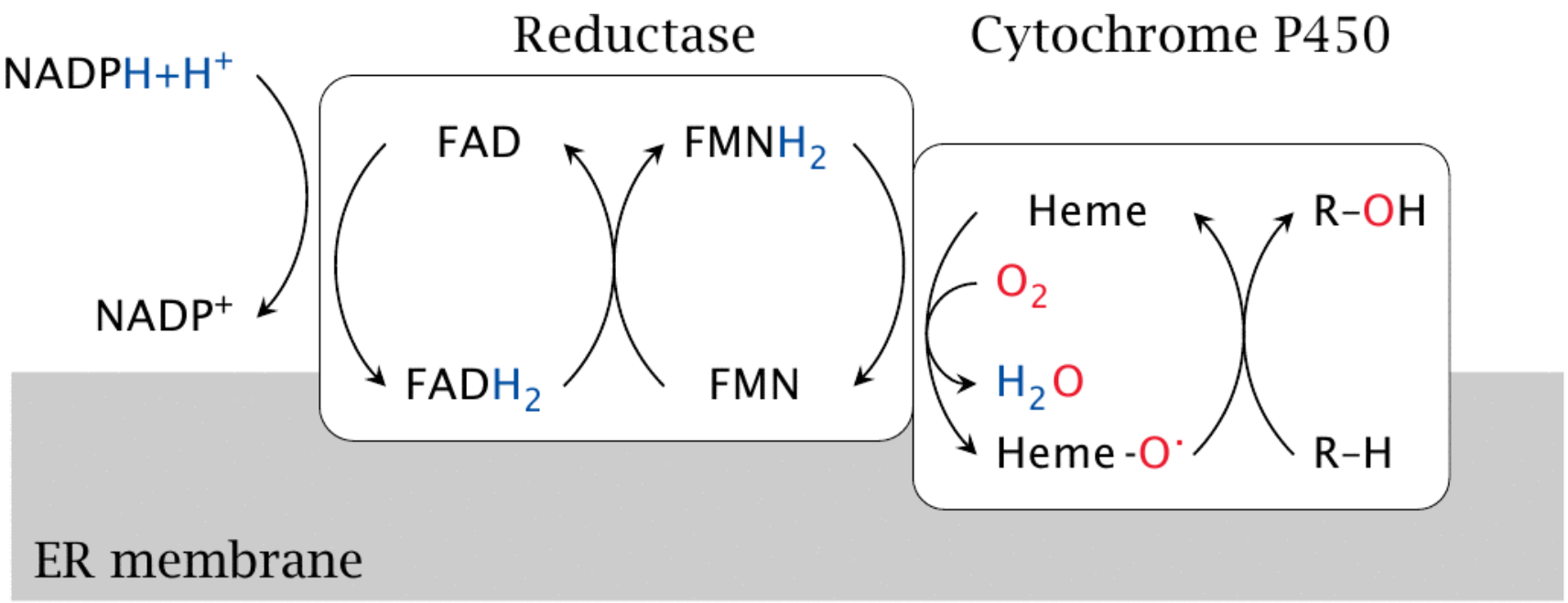
Cost: The acceptable cost depends upon the catalyst lifetime, product value lifetime and product value

Poisoning: Decomposition of catalyst


varies
H2O
H3C
CHCH2
CH3
Fe
N
S
THIOLATE


H2C
CH3
CH2
H3C
N
Fe^III
S
Cys
COOH
COOH

Enzymes
Cytochrome p^{450}



http://watcut.uwaterloo.ca/webnotes/Metabolism/drugsCypActionMode.html

$$R{-}H \xrightarrow{[O]} R{-}OH$$

$$RCH_2{-}OH \xrightarrow{[O]} RCH{=}O + H_2O$$

$$RCH{=}O \xrightarrow{[O]} RCOOH$$

Carbon oxidation

$$R_2N{-}H \xrightarrow{[O]} R_2N{-}OH$$

$$R_3N \xrightarrow{[O]} R_3N{=}O$$

$$R_2S \xrightarrow{[O]} R_2S{=}O$$

Heteroatom oxidation

$$RO{-}CH_2R \xrightarrow{[O]} ROH + O{=}CHR$$

$$R_2N{-}CH_2R \xrightarrow{[O]} R_2NH + O{=}CHR$$

Dealkylation

$$R{-}HC{=}CH{-}R \xrightarrow{[O]} R{-}HC\overset{O}{\triangle}CH{-}R$$

Epoxide formation

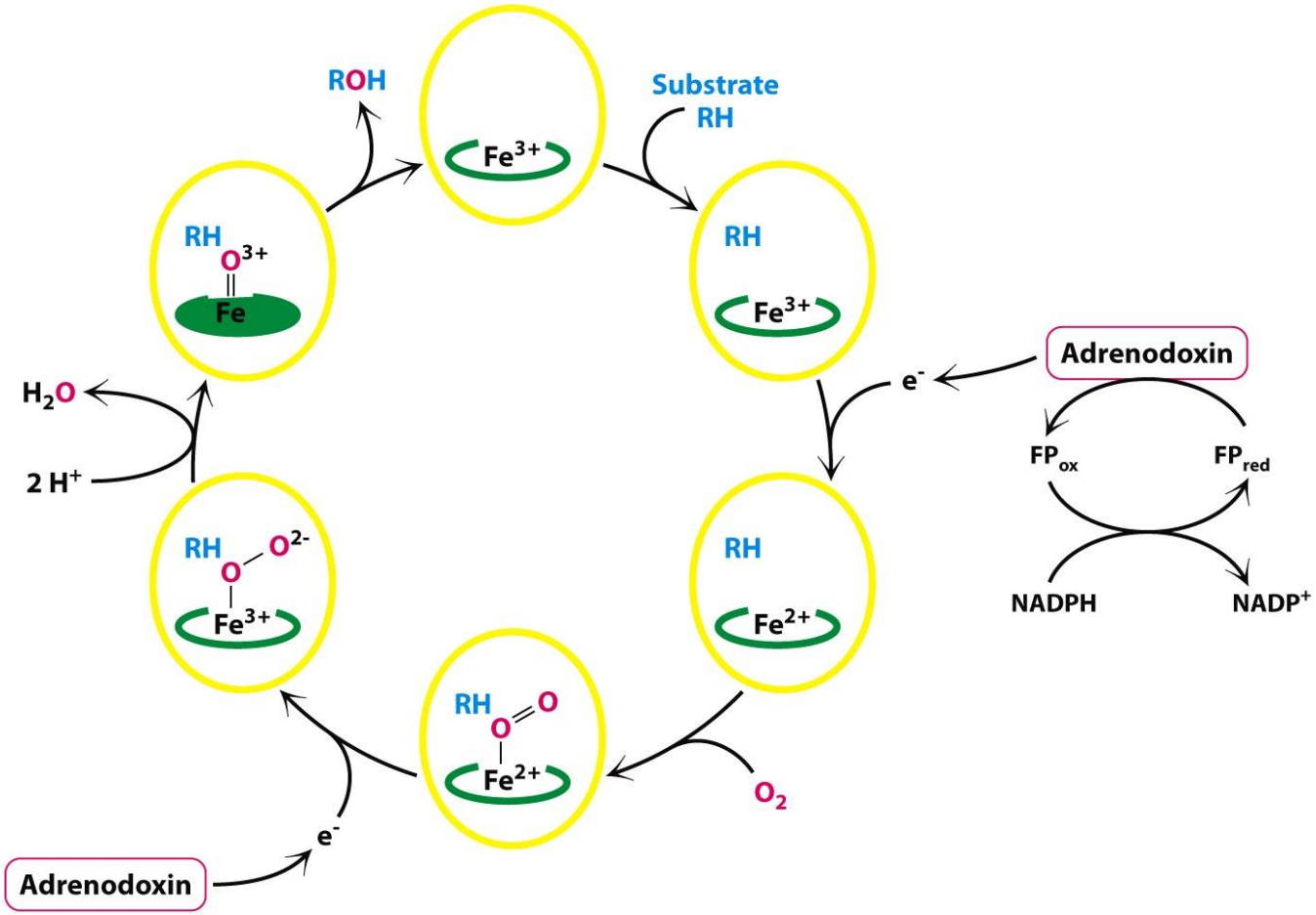


Figure 26.29
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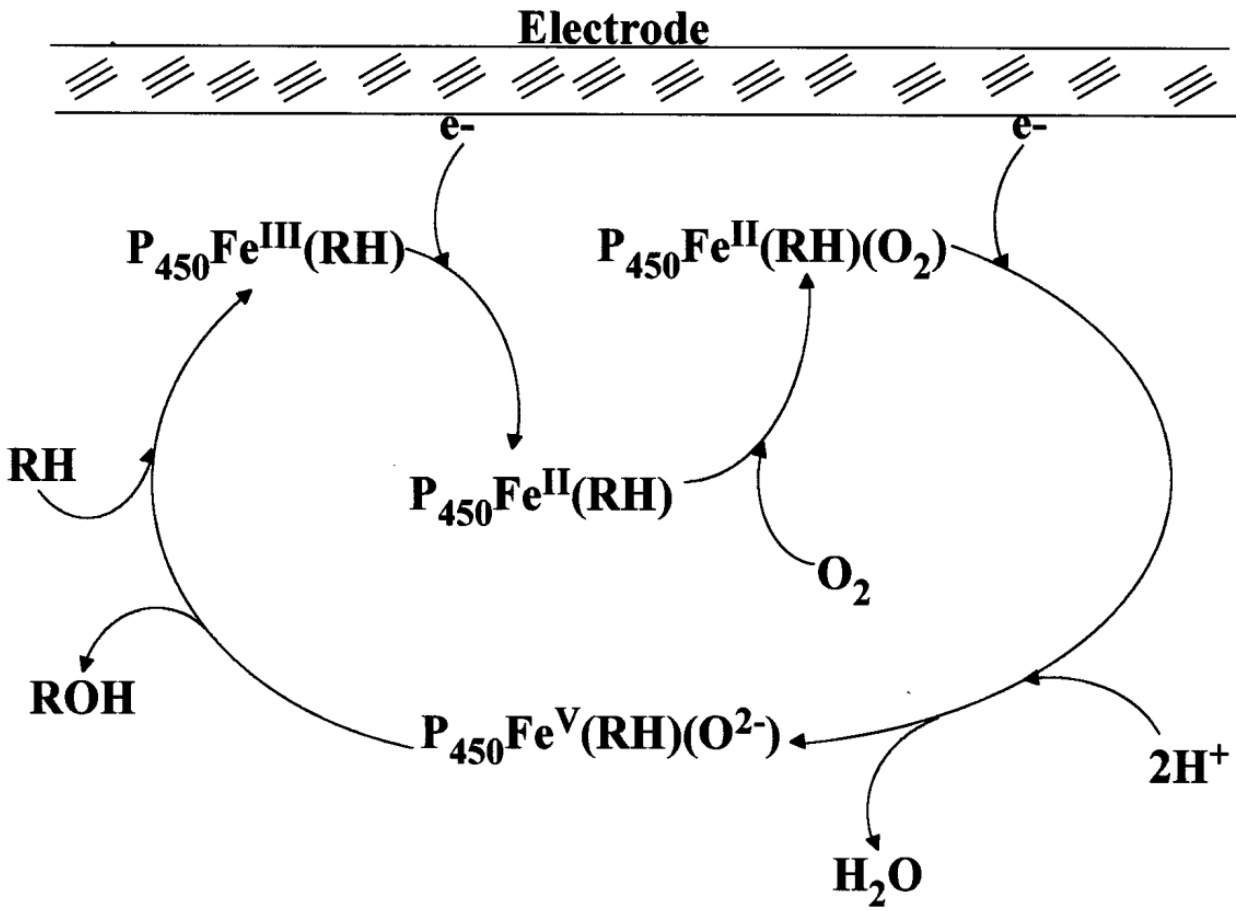

Electrode
e-
e-
$P_{450}Fe^{III}(RH)$
$P_{450}Fe^{II}(RH)(O_2)$
RH
$P_{450}Fe^{II}(RH)$
O_2
ROH
$P_{450}Fe^{V}(RH)(O^{2-})$
$2H^+$
H_2O

Coordination compounds in catalysis
Nobel Prizes

2010 **Heck, Suzuki, Negishi**

2005 **Chauvin, Grubbs, Schrock**

2001 **Knowles, Noyori, Sharpless**

1973 **Wilkinson**

1963 **Ziegler, Natta**

1918 **Haber**

1909 **Ostwald**

Hydrogenation of Unsaturated Hydrocarbons

$$-CH{=}CH- + H_2 \rightarrow -\overset{H}{\overset{|}{C}}H-\overset{H}{\overset{|}{C}}H-$$

$\Delta H = -136\ kJmol^{-1}$; $\Delta G = -101 kJ\ mol^{-1}$

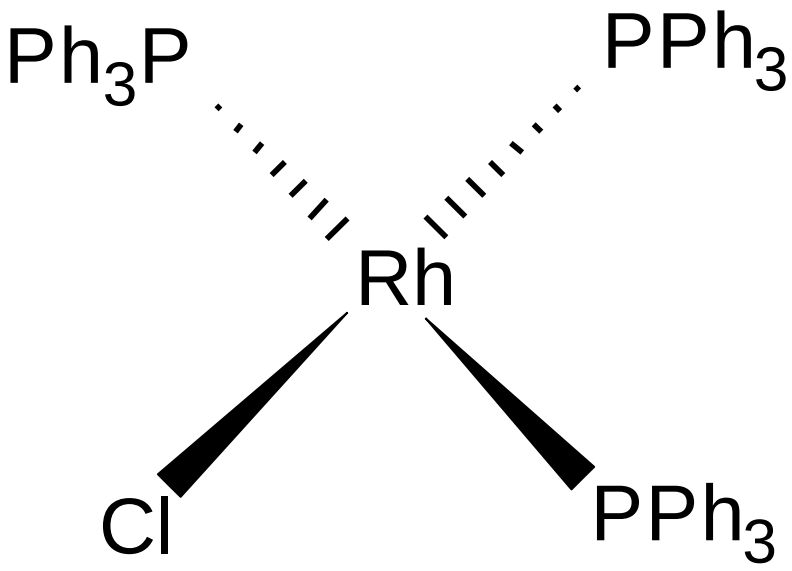
NOBEL : 1973

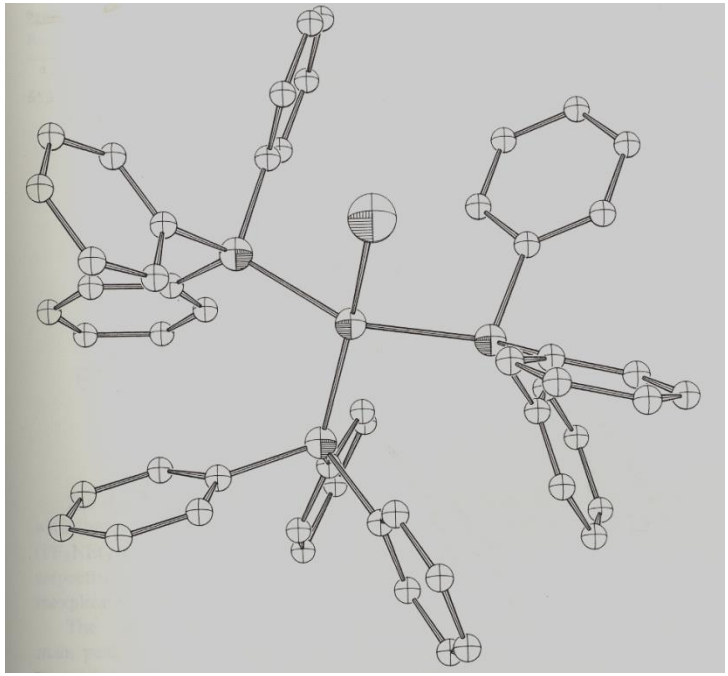
The most common catalyst

$\Downarrow$

Wilkinson's Catalyst, $[RhCl(PPh_3)_3]$

Wilkinson's Catalyst (WC)





Chlorotris(triphenylphosphine)rhodium(I)

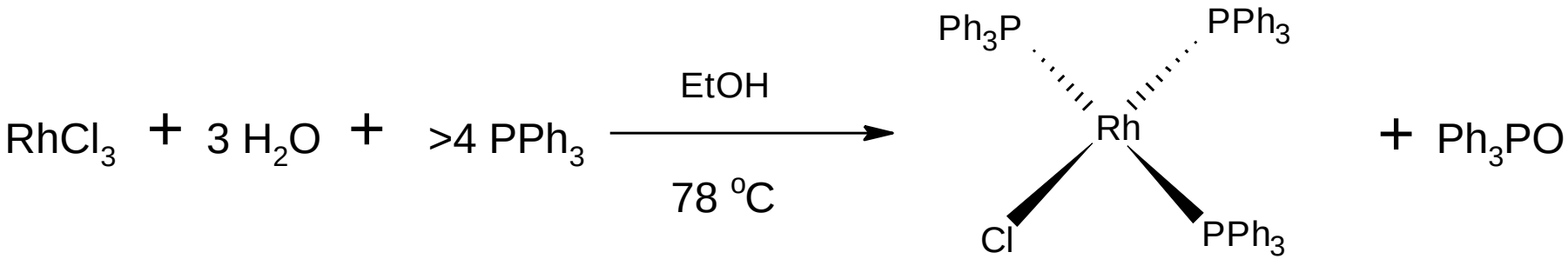
square planar **d^8 configuration**

Geoffrey Wilkinson

- Born July 14, 1921, Yorkshire, England
- Ph.D from Cal Berkeley studying with Glenn Seaborg
- First published WC in 1965 in the Journal *Chemical Communications*
- Nobel Prize in Chemistry 1973 (shared with Ernst Otto Fischer) for their pioneering work, performed independently, on the chemistry of the organometallic, so called sandwich compounds.

Organometallic compounds prepared by Wilkinson in display at Harvard Univ.

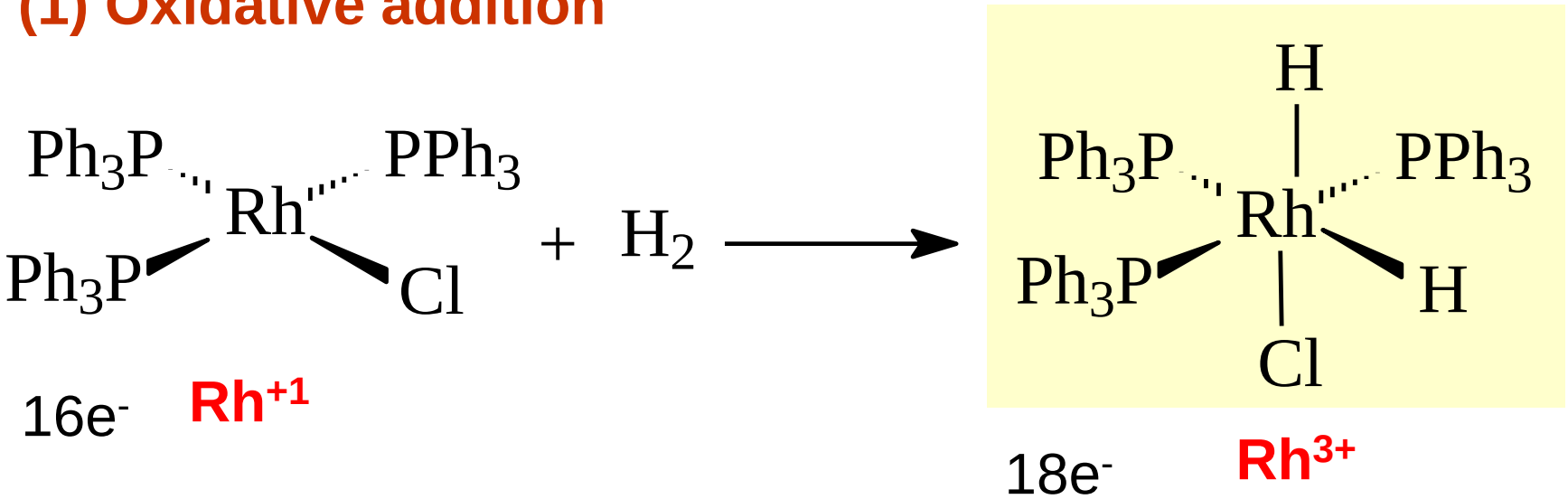
Synthesis of WC



Commercially available

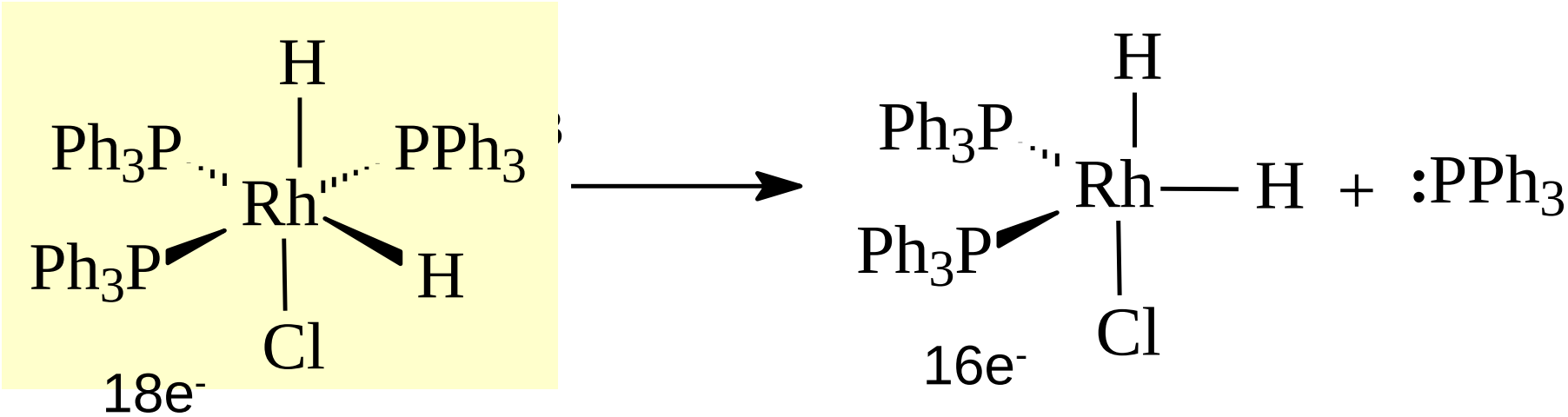
WC in alkene Hydrogenation: Catalytic Steps

(1) Oxidative addition



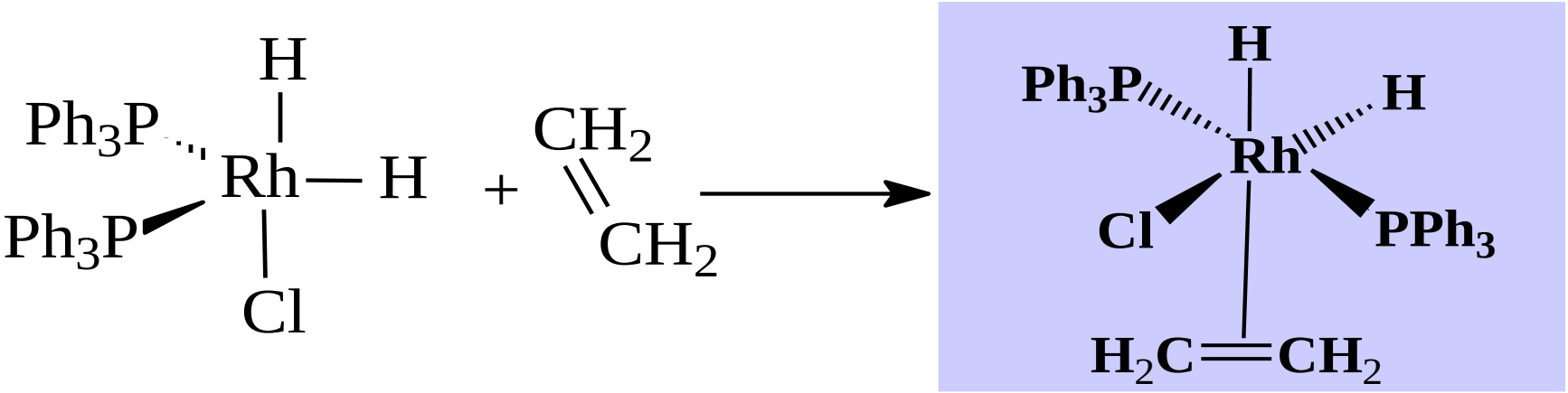
Both the formal oxidation state of the metal and the electron count of the complex increase by two

(2) Ligand Dissociation

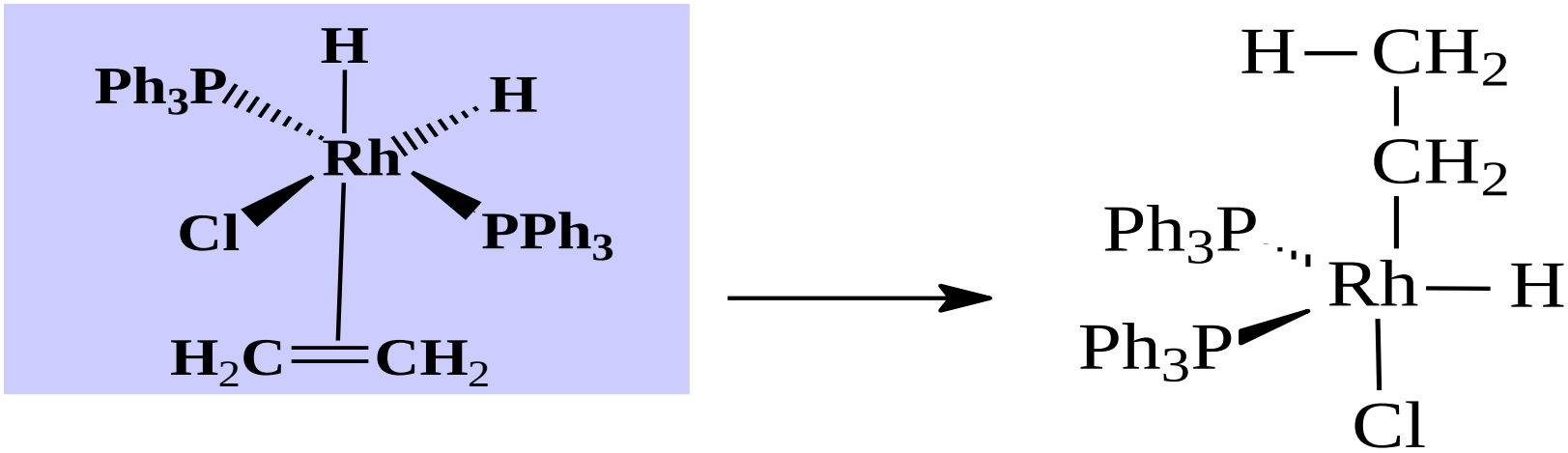


WC in alkene Hydrogenation: Catalytic Steps

(3) Ligand Association

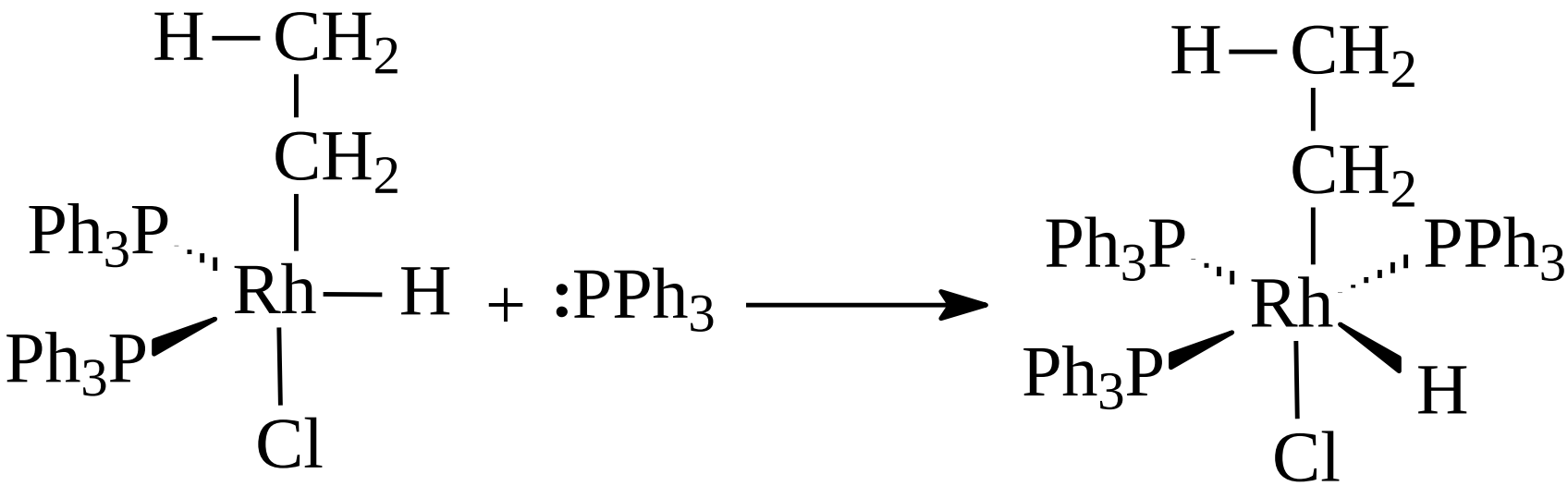


(4) Migration/Insertion



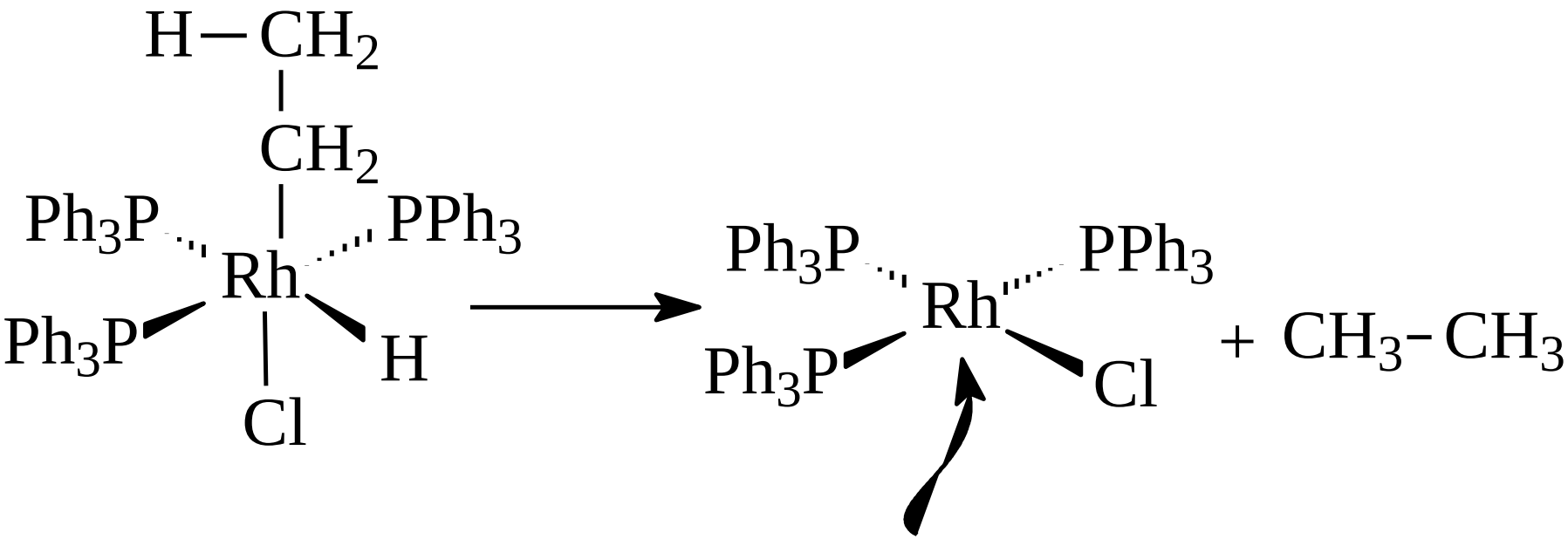
WC in alkene Hydrogenation: Catalytic Steps

(5) Ligand association



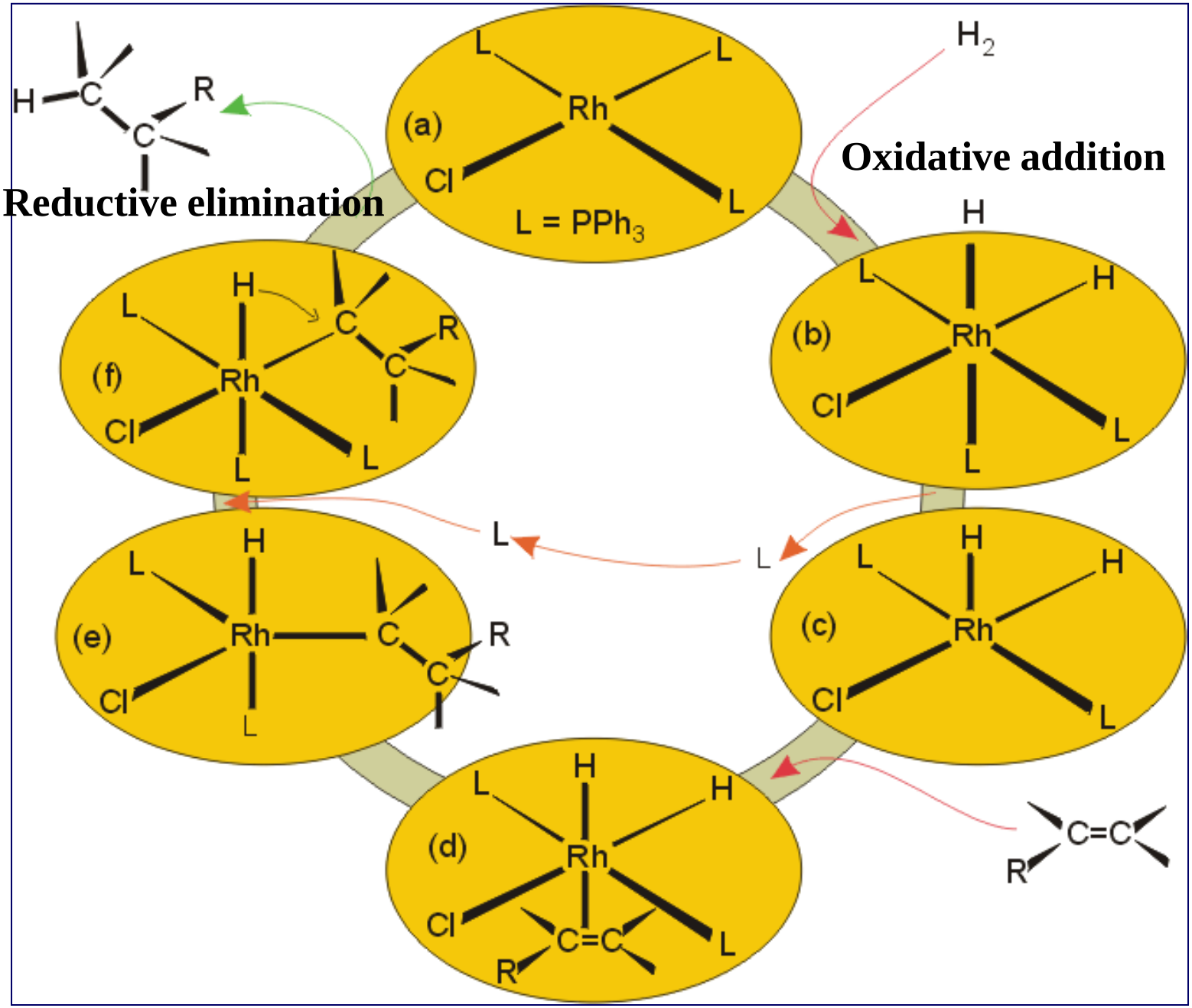
WC in alkene Hydrogenation: Catalytic Steps

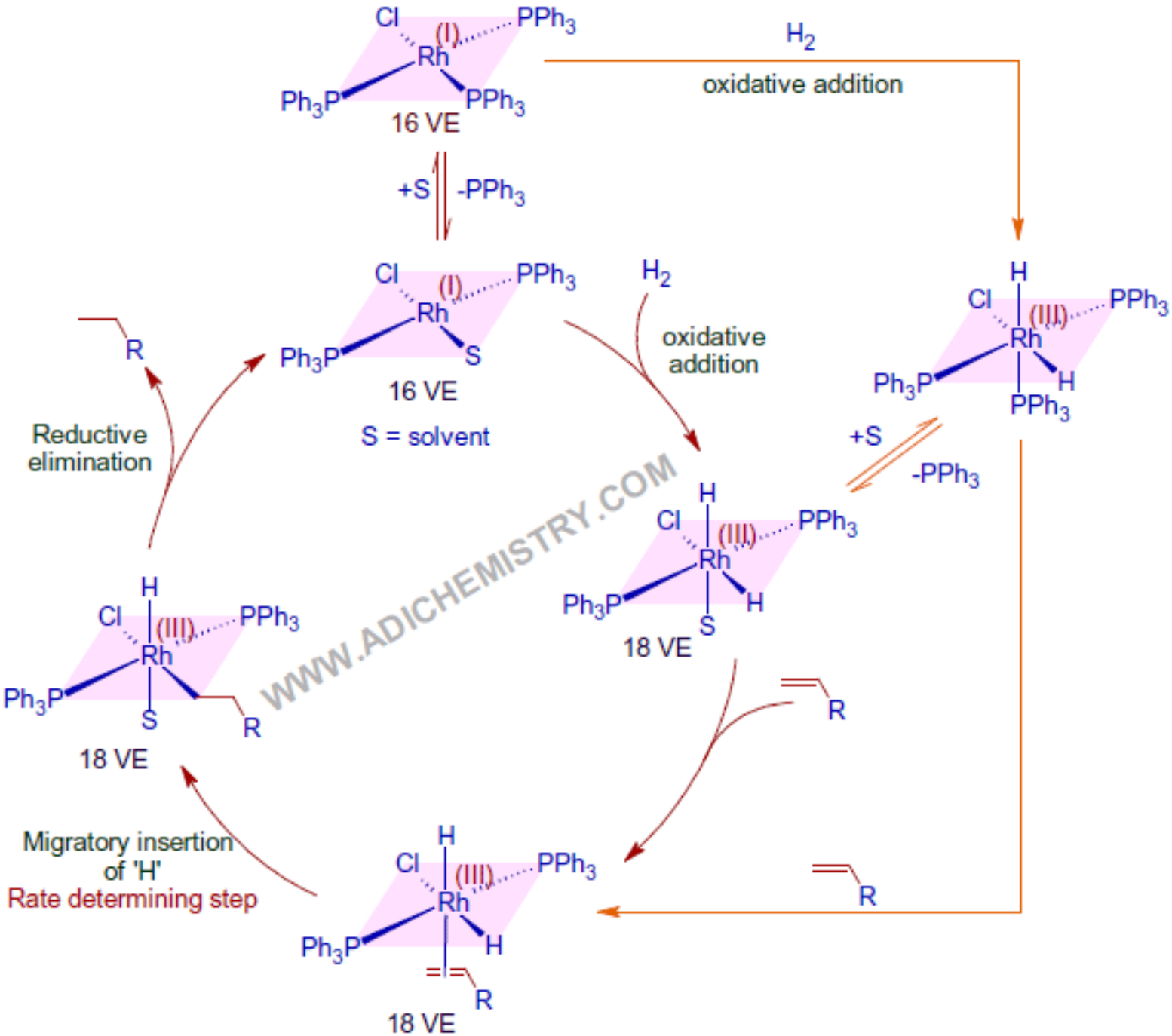
(6) Reductive elimination



(note: regeneration of the catalyst)

WC IN ACTION


Reductive elimination
Oxidative addition
H2
(a)
L = PPh3
(b)
(c)
(d)
(e)
(f)
Rh
Cl
H
L
R
C=C


Cl
(I)
PPh3
Rh
Ph3P
PPh3
16 VE
H2
oxidative addition
+S
-PPh3
Cl
(I)
PPh3
Rh
Ph3P
S
16 VE
S = solvent
H2
oxidative addition
H
Cl
(III)
PPh3
Rh
Ph3P
H
PPh3
+S
-PPh3
H
Cl
(III)
PPh3
Rh
Ph3P
S
H
18 VE
R
R
H
Cl
(III)
PPh3
Rh
Ph3P
H
R
18 VE
Migratory insertion
of 'H'
Rate determining step
H
Cl
(III)
PPh3
Rh
Ph3P
S
R
18 VE
Reductive
elimination
R
WWW.ADICHEMISTRY.COM

WC in alkene Hydrogenation: Additional Notes

Rate of the reaction decreases as the alkyl substitution increases

Highly sensitive to the nature of the phosphine ligand

Analogous complexes with alkylphosphine ligands are inactive

Highly selective for C=C over C=O

Applications

* **Laboratory scale organic synthesis**
* **Production of fine chemicals**

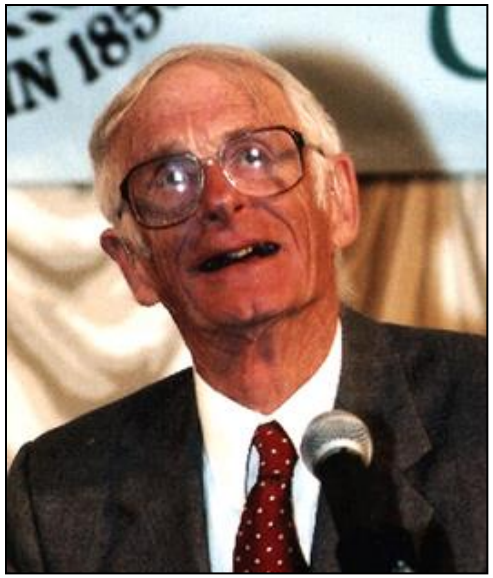
Alkene Hydrogenation & Chirality & Nobel

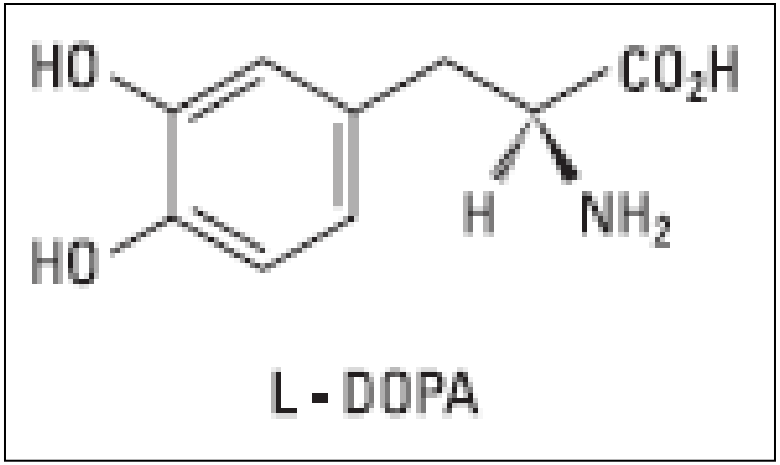
Chiral phosphine ligands have been developed to synthesize optically active products.

Synthesis of L-DOPA (Used in the treatment of Parkinson's diseases)

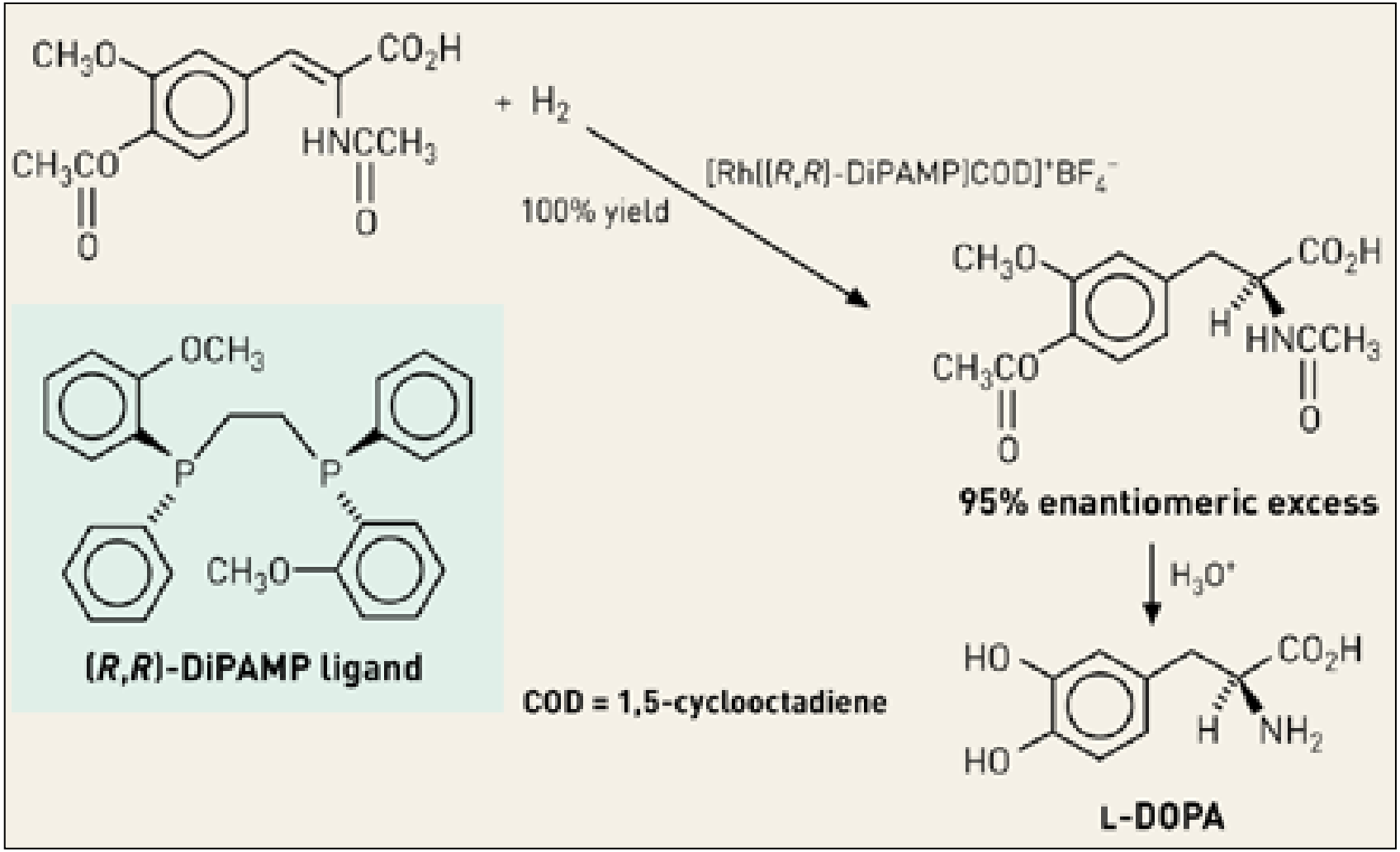
Synthetic route was developed by Knowles & co-workers at Monsanto

Dr. William S. Knowles received Nobel prize in chemistry 2001 along with other two scientists.





Alkene Hydrogenation, Chirality & Nobel



This reaction, developed by Knowles, Vineyard, and Sabacky, was used at Monsanto as a commercial route to the Parkinson's drug L-DOPA.